No Interior First Contact for the Stationary and Vertex-Average Trajectories in Markovian Persuasion

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Abstract

In a Markovian persuasion game with irreducible transition matrix M, let π_M be the invariant distribution and let

$$\Phi(\delta) = v_\delta(\pi_M), \qquad \Psi(\delta) = \sum_{i \in K} \pi_M(i) v_\delta(e_i), \qquad \delta \in [0, 1),$$

where v_δ is the normalized δ-discounted value. Shaiderman's comparison of the monotone trajectories Φ and Ψ leaves open whether there can be an interior first contact point $\delta_0 \in (0, 1)$ such that $\Phi > \Psi$ below δ_0 and $\Phi = \Psi$ from δ_0 onward. We show that this cannot occur. More precisely, equality at a single discount factor already implies equality at every discount factor. Equivalently, the equality set is either empty or all of $[0, 1)$. The proof is elementary: equality in the Jensen inequality at the full-support invariant distribution forces the value function to be affine on the whole simplex; the Bellman equation then implies that the one-period payoff is dominated by its affine interpolation from the vertices, and full revelation is therefore optimal for every discount factor.

1 Model and statement

This note concerns the Markovian persuasion model studied by Lehrer and Shaiderman [2] and the comparison of the two value trajectories in Shaiderman [3]. The underlying one-shot Bayesian persuasion problem goes back to Kamenica and Gentzkow [1].

Let $K = \{1, \ldots, k\}$ be a finite state space and let $\Delta(K)$ be the probability simplex. Vectors in $\Delta(K)$ are treated as row vectors. Let M be an irreducible stochastic matrix on K, and denote its unique invariant distribution by $\pi = \pi_M$. Irreducibility implies that $\pi_i > 0$ for every $i \in K$.

The stage payoff induced by a posterior belief $p \in \Delta(K)$ is denoted by $u(p)$. In the usual Markovian persuasion notation,

$$u(p) = \sum_{i \in K} p_i V(i, b(p)),$$

where $b(p)$ is the receiver's tie-broken best response to belief p. Let $v_\delta(p)$ be the sender's normalized discounted value from initial prior p:

$$v_\delta(p) = \sup_\sigma \mathbb{E}_{p,\sigma} \left[(1 - \delta) \sum_{n \geq 1} \delta^{n-1} u(p_n) \right], \qquad \delta \in [0, 1),$$

where p_n is the receiver's posterior at stage n. We assume the standard signal-cardinality condition under which the sender can fully reveal the current state; for instance, a signal set of cardinality at least $|K|$ is enough.

The dynamic programming formula is

$$v_\delta(p) = \sup_{\mu \in \mathcal{S}(p)} \int_{\Delta(K)} \Big((1-\delta)u(q) + \delta v_\delta(qM) \Big) \, d\mu(q). \tag{1}$$

where $\mathcal{S}(p)$ is the set of finite-support probability measures on $\Delta(K)$ with barycenter p. Equivalently,

$$v_\delta = \operatorname{cav} \Big((1-\delta)u + \delta \, v_\delta \circ M \Big),$$

so v_δ is concave on $\Delta(K)$.

Define

$$\Phi(\delta) := v_\delta(\pi), \qquad \Psi(\delta) := \sum_{i \in K} \pi_i v_\delta(e_i).$$

Since v_δ is concave and $\pi = \sum_i \pi_i e_i$, one always has $\Phi(\delta) \geq \Psi(\delta)$. The purpose of this note is to prove that the set of discount factors at which equality holds has no interior threshold.

Theorem 1 (No interior first contact)**.** *For every Markovian persuasion game with irreducible transition matrix M, exactly one of the following alternatives holds:*

$$\Phi(\delta) = \Psi(\delta) \quad \textit{for every } \delta \in [0,1),$$

or

$$\Phi(\delta) > \Psi(\delta) \quad \textit{for every } \delta \in [0,1).$$

Consequently there is no example in which equality first occurs at an interior point $\delta_0 \in (0,1)$.

We prove the theorem through the following slightly sharper characterization. Let

$$\ell_u(p) := \sum_{i \in K} p_i u(e_i)$$

be the affine interpolation of the one-period payoff from the vertices of the simplex.

Proposition 1 (Characterization of equality)**.** *The following are equivalent:*

(i) $\Phi(\delta) = \Psi(\delta)$ *for some* $\delta \in [0,1)$*;*

(ii) $\Phi(\delta) = \Psi(\delta)$ *for every* $\delta \in [0,1)$*;*

(iii) $u(p) \leq \ell_u(p)$ *for every* $p \in \Delta(K)$*.*

When these conditions hold, full revelation is optimal for every initial prior and every discount factor, and

$$v_\delta(p) = (1-\delta) \sum_{n \geq 1} \delta^{n-1} \ell_u(pM^{n-1}). \tag{2}$$

2 Proof

We first record the elementary convex-analytic fact that will be used at the invariant distribution.

Lemma 1 (Equality at a full-support mixture forces affinity). *Let $f : \Delta(K) \to \mathbb{R}$ be concave, and let $\pi \in \Delta(K)$ have full support. If*

$$f(\pi) = \sum_{i\in K} \pi_i f(e_i),$$

then

$$f(p) = \sum_{i\in K} p_i f(e_i) \qquad \textit{for every } p \in \Delta(K).$$

Proof. Set $L(p) := \sum_i p_i f(e_i)$. Concavity gives $f(p) \geq L(p)$ for all $p \in \Delta(K)$. Fix $p \in \Delta(K)$. Since π has full support, choose $\varepsilon > 0$ small enough that

$$r := \frac{\pi - \varepsilon p}{1 - \varepsilon} \in \Delta(K).$$

Then $\pi = \varepsilon p + (1 - \varepsilon)r$. Hence

$$\begin{aligned} L(\pi) &= f(\pi) \\ &\geq \varepsilon f(p) + (1-\varepsilon) f(r) \\ &\geq \varepsilon L(p) + (1-\varepsilon) L(r) \\ &= L(\pi). \end{aligned}$$

All inequalities are equalities, so $f(p) = L(p)$. Since p was arbitrary, the claim follows. □

Lemma 2 (One equality implies vertex domination of the stage payoff). *If $\Phi(\bar\delta) = \Psi(\bar\delta)$ for some $\bar\delta \in [0,1)$, then*

$$u(p) \leq \ell_u(p) \qquad \textit{for every } p \in \Delta(K).$$

Proof. Assume $\Phi(\bar\delta) = \Psi(\bar\delta)$. By Lemma 1, applied to $f = v_{\bar\delta}$, the value function $v_{\bar\delta}$ is affine on $\Delta(K)$. Write

$$L(p) := v_{\bar\delta}(p) = \sum_i p_i v_{\bar\delta}(e_i).$$

At a vertex e_i, every feasible split with barycenter e_i is degenerate at e_i. Therefore the Bellman equation gives

$$L(e_i) = (1-\bar\delta) u(e_i) + \bar\delta L(e_i M). \tag{3}$$

Multiplying (3) by p_i and summing over i, we get

$$L(p) = (1-\bar\delta)\ell_u(p) + \bar\delta L(pM) \qquad \text{for every } p \in \Delta(K). \tag{4}$$

On the other hand, the degenerate split of p into itself is feasible in the Bellman equation. Thus

$$L(p) = v_{\bar\delta}(p) \geq (1-\bar\delta) u(p) + \bar\delta L(pM). \tag{5}$$

Subtracting (5) from (4) and using $\bar\delta < 1$ yields $u(p) \leq \ell_u(p)$. □

Lemma 3 (Vertex domination makes full revelation optimal)**.** *If $u \le \ell_u$ on $\Delta(K)$, then for every $\delta \in [0,1)$ and every $p \in \Delta(K)$,*

$$v_\delta(p) = (1-\delta)\sum_{n\ge 1} \delta^{n-1} \ell_u(pM^{n-1}).$$

In particular, v_δ is affine in p for every δ.

Proof. Fix $p \in \Delta(K)$, $\delta \in [0,1)$, and a signaling policy σ. Let p_n be the posterior at stage n. Since p_n is the conditional distribution of X_n given the receiver's information, the law of total expectation gives

$$\mathbb{E}_{p,\sigma}[p_n] = \mathcal{L}_p(X_n) = pM^{n-1}. \tag{6}$$

Therefore, using $u \le \ell_u$ and the affinity of ℓ_u,

$$\begin{aligned}
\mathbb{E}_{p,\sigma}\left[(1-\delta)\sum_{n\ge 1}\delta^{n-1}u(p_n)\right] &\le (1-\delta)\sum_{n\ge 1}\delta^{n-1}\mathbb{E}_{p,\sigma}[\ell_u(p_n)] \\
&= (1-\delta)\sum_{n\ge 1}\delta^{n-1}\ell_u(\mathbb{E}_{p,\sigma}[p_n]) \\
&= (1-\delta)\sum_{n\ge 1}\delta^{n-1}\ell_u(pM^{n-1}).
\end{aligned}$$

Taking the supremum over σ gives the same upper bound for $v_\delta(p)$.

The bound is attained by full revelation: at every stage the sender sends a signal identifying the realized state X_n. Then $p_n = e_{X_n}$ and hence $u(p_n) = u(e_{X_n}) = \ell_u(e_{X_n})$. Since X_n has distribution pM^{n-1}, the expected payoff from this policy is exactly

$$(1-\delta)\sum_{n\ge 1}\delta^{n-1}\ell_u(pM^{n-1}).$$

Thus the upper bound is tight, proving the formula. The right-hand side is affine in p, so v_δ is affine in p. □

Proof of Proposition 1. Condition (ii) trivially implies condition (i). Lemma 2 shows that (i) implies (iii). Lemma 3 shows that (iii) implies that every v_δ is affine. Hence for every $\delta \in [0,1)$,

$$\Phi(\delta) = v_\delta(\pi) = \sum_i \pi_i v_\delta(e_i) = \Psi(\delta),$$

which is condition (ii). The formula (2) and the optimality of full revelation are exactly Lemma 3. □

Proof of Theorem 1. Concavity of v_δ gives $\Phi(\delta) \ge \Psi(\delta)$ for every $\delta \in [0,1)$. If equality holds for at least one δ, then Proposition 1 implies equality for every δ. If equality holds for no δ, then the inequality is strict for every δ. These are the two alternatives in the theorem, and neither admits an interior first equality point. □

3 Discussion

The proof uses neither the monotonicity of Φ nor the monotonicity of Ψ. Those monotonicities motivate the open question, but the obstruction to an interior first contact is more basic. At any equality point, Jensen's inequality for the concave function v_δ is tight at the full-support mixture $\pi = \sum_i \pi_i e_i$. This is already strong enough to make v_δ affine on the whole simplex. The Bellman equation then transfers this affinity back to the one-period payoff and yields $u \le \ell_u$. Once this pointwise domination is known, no dynamic information policy can beat full revelation, because the posterior process always has mean pM^{n-1} at stage n.

Thus the possible "Case B" in which the two graphs meet and then coincide can only occur with threshold $\delta_0 = 0$, i.e., in the degenerate sense that the two trajectories coincide for all discount factors. A genuinely interior threshold $\delta_0 \in (0, 1)$ is impossible.

References

[1] Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.

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[3] Dimitry Shaiderman. On the monotonicity and rate of convergence of the Markovian persuasion value. arXiv:2512.06794, 2025.